

# Correspondence Principle and the Limits of Quantum Mechanical Systems

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## Abstract

This essay discusses the limits where a quantum system can be treated in accordance with classical theory. It explores the transition of harmonic quantum systems into classical systems and the scales at which the classical interpretation holds.

## 1 Correspondence Principle

The correspondence principle is the concept that quantum and classical theories must be continuous; that is, quantum and classical behavior must converge at sufficiently large scales. First introduced by Max Planck in 1900, the correspondence principle is a cornerstone of quantum and statistical mechanics, alongside Ehrenfest's theorem linking the behavior of quantum systems and macroscopic phenomena. In the early development of quantum physics, it was believed that quantum and classical mechanics were completely separate. However, this perspective led to conceptual contradictions regarding how the transition between the two occurs. One of the central questions that arose was where the boundary between quantum and classical mechanics lies. Modern physics recognizes that there is no sharp boundary; rather, quantum mechanics gradually reduces to classical mechanics under appropriate conditions.

Although there is no dividing line distinguishing quantum and classical behavior, the distinction remains highly useful, allowing us to neglect quantum effects in macroscopic systems and focus on the general behavior of complex phenomena.

### 1.1 Characteristics of Quantum and Classical Systems

One of the primary distinctions between quantum and classical systems lies in the quantization of energy, which was the historical motivation for the development of quantum mechanics. In classical theory, a system's energy is continuous and can assume any value. In contrast, quantum mechanics imposes discrete energy levels. Another crucial difference is the probabilistic nature of quantum mechanics, as demonstrated by Heisenberg's uncertainty principle.

Despite these differences, many parallels exist between quantum and classical mechanics. Most concepts have analogous roles, albeit with subtle differences.

## 2 Harmonic Oscillators

### 2.1 Classical Oscillators

The harmonic oscillator has a clear quantum analogue and is highly interesting for this discussion as it can be used in many circumstances as a low-order approximation of more complex systems. Classically, the harmonic oscillator is characterized by a restoring force proportional to the displacement of a particle:

$$F(x) = -kx$$

where  $k$  is the restoration constant, which is related to the oscillator's mass and angular frequency by  $\omega^2 = k/m$ . With this, we can easily write the potential energy created by this force, arriving at:

$$V(x) = \frac{1}{2}m\omega^2x^2$$

Assuming that this is the only force acting on the oscillator, we can then write its total energy by adding the kinetic energy to the potential:

$$E_T = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2$$

The total energy is also equal to the maximum potential energy. If the oscillation has amplitude  $A$ , then the total energy is:

$$E_T = \frac{1}{2}m\omega^2 A^2$$

As previously discussed, a quantum system operates in a probabilistic manner, so it is more useful if we analyze the probability of finding our classical particle in any given position. The probability density of a generic classical oscillator is defined by its total and potential energy, as seen in [1]:

$$P(x) = \frac{\omega}{2\pi} \sqrt{\frac{2m}{E_T - V}} \quad (1)$$

Substituting the values for  $E_T$  and  $V$  into equation (1), we arrive at the probability function:

$$P(x) = \frac{1}{\pi} \frac{1}{\sqrt{A^2 - x^2}}, \quad |x| \leq A. \quad (2)$$

Plotting this, we can observe that the probability of finding the mass grows as it approaches the points of maximum potential energy, i.e. the points of minimal velocity, the region in which the particle spends the most time.

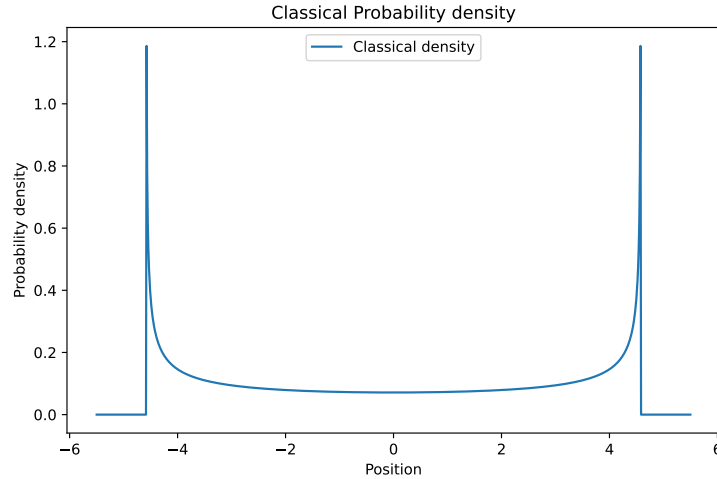


Figure 1: Probability density of the classical harmonic oscillator.

Having understood the classical oscillator, we now turn to its quantum analogue, where similar equations reveal strikingly different behavior.

## 2.2 Quantum Oscillators

In the quantum regime, the main difference is that now we deal with operators instead of straight forward variables, the Hamiltonian operator resembles closely the classical Hamiltonian:

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2$$

Taking  $\psi(x)$  as an eigenfunction of the Hamiltonian operator and  $\hat{p} = -i\hbar\frac{d}{dx}$ , the time-independent Schrödinger's equation becomes:

$$E\psi(x) = -\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + \frac{1}{2}m\omega^2x^2\psi(x)$$

The solution for this equation is non trivial and can be found in [2], the wave function is:

$$\psi_n(x) = \frac{1}{\sqrt{2^n n!}} \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\frac{m\omega x^2}{2\hbar}} H_n\left(\sqrt{\frac{m\omega}{2\hbar}}x\right) \quad (3)$$

where  $n = 0, 1, 2, \dots$  labels the quantized energy levels  $E_n = \hbar\omega(n + 1/2)$ , and  $H_n$  are Hermite polynomials. With this we can calculate the probability density using  $P(x) = |\psi(x)|^2$ . Plotting this for multiple energy levels, we arrive at the following graphs:

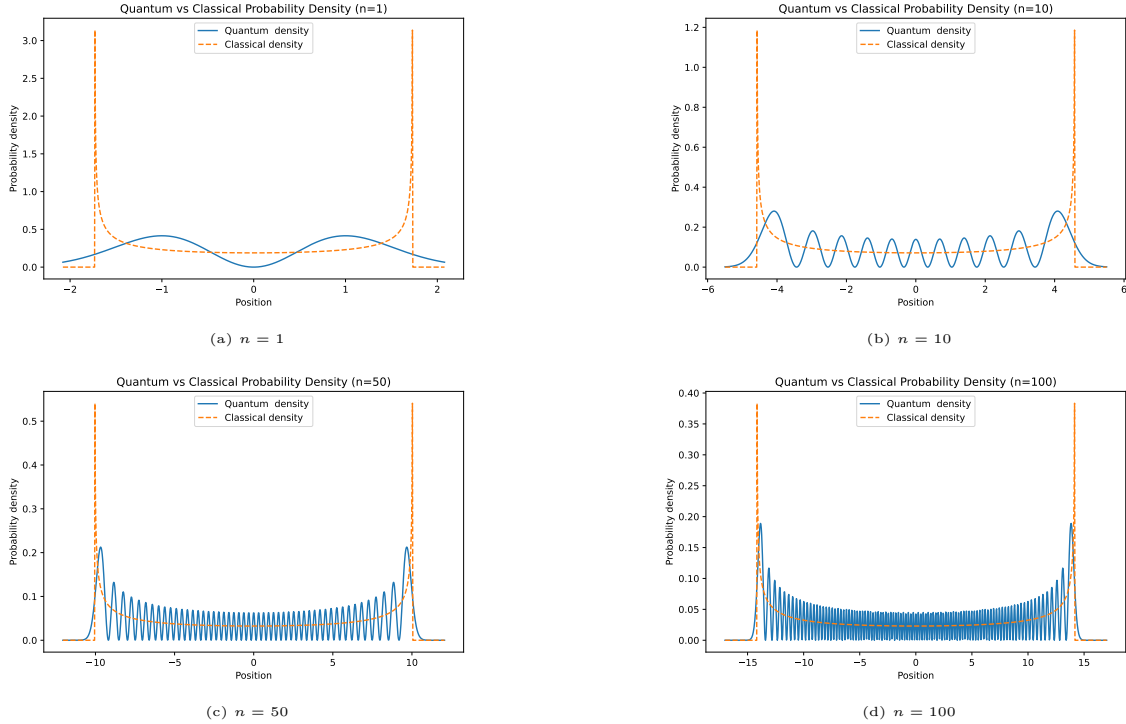


Figure 2: Probability densities of the quantum harmonic oscillator for different energy levels.

We can see that the quantum model appears to approach the classical one, only instead having rapid oscillations in the probability density.

Using the following function to calculate the relative difference between the models:

$$\Delta(n) = \int_{\mathbb{R}} |P_c(x) - P_q(x, n)| dx$$

We arrive at the graph showing how the classical and quantum models differ in relation to  $n$ :

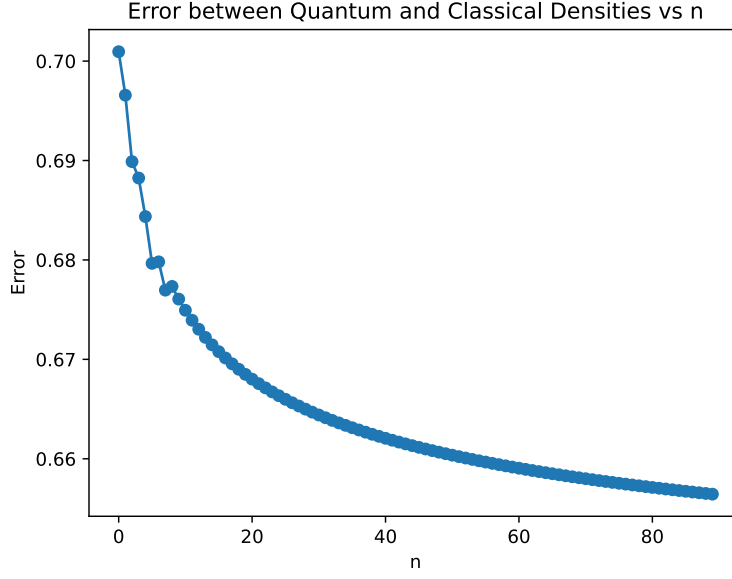


Figure 3: Relative difference  $\Delta(n)$  between quantum and classical probability densities as a function of  $n$ .

With this we can clearly see that the error decreases as  $n$  increases, suggesting that at sufficiently large  $n$ , the models will agree. By analyzing the energy of both systems, one can find at which energy level the models are equivalent. The energies of the quantum and classical systems are:

$$E_q = \hbar\omega \left( n + \frac{1}{2} \right) \quad \& \quad E_c = \frac{1}{2} m\omega^2 A^2$$

Equating the energies we can express the amplitude as a function of  $n$ :

$$A = \sqrt{\frac{2E_q}{m\omega^2}} = \sqrt{\frac{\hbar}{m\omega}} (2n + 1) \quad (4)$$

For a classical system  $A \gg \sqrt{\frac{\hbar}{m\omega}}$ , suggesting that for a large energy  $E_q$ , the quantum system will behave in accordance with the classical picture.

### 3 Limits of Classical Theory

One of the characteristics of a classical system is the fact that the energy appears to be continuous, that is, the energy can assume any arbitrary value within a meaningful scale for the problem. So if the spacing between the allowed energy levels is small enough, we can analyze the relative energy difference between two adjacent quantum states:

$$\frac{E_{n+1} - E_n}{E_{n+1}} = \frac{\hbar\omega(n+1-n)}{\hbar\omega(n+1+1/2)} = \frac{2}{2n+3}$$

If we require that the difference is equal to any arbitrary error constant  $\epsilon$ , we can calculate the required energy of the system such that the quantum and classical interpretations are within a difference of  $\epsilon$ . Plugging in the values, we arrive at:

$$n = \frac{1}{\epsilon} - \frac{3}{2} \approx \frac{1}{\epsilon} \quad \Rightarrow \quad E \approx \frac{\hbar\omega}{\epsilon} \quad (5)$$

This shows that the energy at which we can consider the oscillator classical depends linearly on  $\omega$ . A small angular frequency means that the oscillator can be seen as a stationary particle on short time scales, a system that can clearly be explained by classical theory. Therefore, the separation between classical and quantum mechanics is not dependent on energy alone.

### 3.1 Quantum Action Criterion

To refine our understanding, let us turn to the concept of the action variable. The action variable [3] is defined as:

$$I = \frac{1}{2\pi} \oint_{\Omega} p(x) dx, \quad (6)$$

that is, the integral of momentum over one full cycle of the trajectory in phase space. Let us compute the action variable for our system. Rewriting  $p$  as a function of  $x$ , we arrive at:

$$p(x) = \sqrt{m(2E - m\omega^2 x^2)}$$

Calculating this integral in the interval  $t \in [0, T]$ , we arrive at:

$$I = \frac{ET}{2\pi} = \frac{E}{\omega}$$

So we can rewrite the expression found in equation (5) as:

$$I \approx \frac{\hbar}{\epsilon} \quad (7)$$

That shows that for a quantum and classical system to be equivalent within an error window of  $\epsilon$ , their respective action variables need to be of order  $\hbar/\epsilon$ . Let us see an example.

For a very small  $\epsilon = 10^{-5}$ , a system with action:

$$I \approx 10^{-29} Js$$

could be interpreted as a classical system within a 0.001% error margin. To put that into perspective, the action variable of a quantum system such as an electron orbiting the nucleus is of order  $10^{-34} Js$  [4], many orders of magnitude smaller than the value we obtained.

## 4 Conclusion

The distinction between classical and quantum behavior does not depend solely on the system's energy but on its action variable. According to the action criterion, when  $I \gg \hbar$  the system can be viewed as classical, more precisely, the error between the quantum and classical descriptions can be estimated as  $\epsilon \approx \hbar/I$ . Systems with an action greater than  $10^{-29} Js$  will exhibit negligible quantum effects.

## References

- [1] David J. Griffiths and Darrell F. Schroeter. *Introduction to Quantum Mechanics*. Cambridge University Press, 3rd edition, 2018.
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- [4] Niels Bohr. On the constitution of atoms and molecules. *Philosophical Magazine*, 26:1–25, 1913.